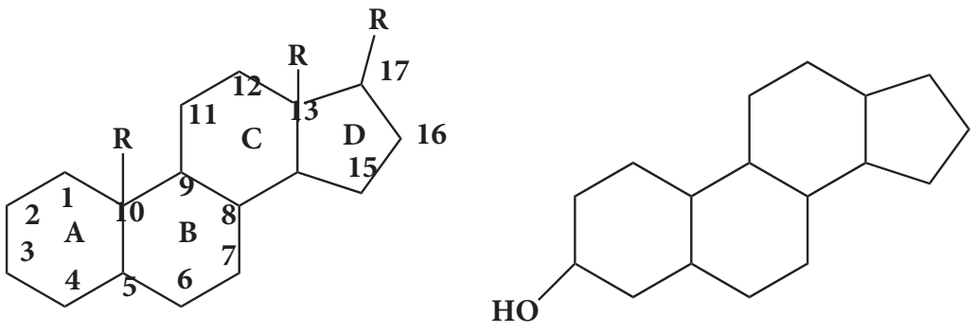
6.5 Derived Lipids:

6.5.1. Steroids:

Steroids are derived lipids. They may be recognized by their cyclo pentane perhydro phenatherene ring-tetracyclic skeleton. It consists of three six-membered ring (A,B,C) and one five-membered ring(D), as shown in the diagram.



The Steroid Carbon Skeleton **General structure of sterol**

Figure 6.16 Carbon skeleton of steroids and sterols

6.5. 2 Sterols

Sterols, also known as steroid alcohols, are a subgroup of steroids with a hydroxyl group at the 3-position of the A-ring. They are amphipathic lipids as the hydroxyl group on the A ring is polar and the rest of the aliphatic chain is non-polar.

Types:

- Phytosterols (Example: Stigmasterol)
- Zoosterols (Example: Cholesterol)

Sterols of plants are called phytosterols and sterols of animals are called zoosterols. The most important zoosterol is cholesterol; phytosterols include campesterol, sitosterol and stigmasterol. Ergosterol is a sterol present in the cell membrane of fungi, where it serves a role similar to cholesterol in animal cells.

Note

Phytosterols as a nutritional supplement:

It have been shown in clinical trials to block cholesterol absorption sites in the human intestine, thus helping to reduce cholesterol absorption in humans. At present, the American Heart Association has recommended that supplemental plant sterols be taken only by those diagnosed with elevated cholesterol, as phytosterols can interfer with cholesterol absorption.

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